

Question 1

(d) [Non-Cooking]

(i) Comparing Monte Carlo and Importance Sampling

Monte Carlo:

$$g(x) = e^{\cos(x)} - x \text{ under } X \sim \text{Exp}(1)$$

$$E[g(X)] \approx \frac{1}{n} \sum_{i=1}^n g(x_i) \approx 1.150$$

Variance:

$$V[g(x)] \approx \frac{1}{n-1} \sum_{i=1}^n (g(x_i) - \bar{g})^2$$

$$V[g(x)] \approx 0.02$$

Importance Sampling: $Y \sim \text{Exp}(2)$

$$E[g(X)] \approx \frac{1}{n} \sum_{i=1}^n g(y_i) \cdot \frac{p_X(y_i)}{p_Y(y_i)}$$

$$p_X(y_i) = \lambda e^{-\lambda y_i} \text{ for Exp}(1) \text{ where } \lambda=1$$

$$p_Y(y_i) = \lambda e^{-\lambda y_i} \text{ for Exp}(2) \text{ where } \lambda=2$$

$$E[g(X)] \approx 1.165$$

Variance:

$$V[g(X)] \approx \frac{1}{n-1} \sum_{i=1}^n \left(g(y_i) \cdot \frac{p_X(y_i)}{p_Y(y_i)} - \bar{g} \right)^2$$

$$V[g(X)] \approx 1.01$$

This calculation shows that importance sampling is more effective in this context which is closer to the ground truth 1.160.

(ii) Comparing the Three Plots

$$X \sim \text{Exp}(1): f(x) = \lambda e^{-\lambda x}, x \geq 0, \lambda = 1$$

$$X \sim \text{Exp}(2): g(x) = 2e^{-2x}, x \geq 0$$

Target function

$$f(x)p(x) = e^{\cos(x)-2x}$$

$$\text{Exp}(1) \text{ PDF: } p(x) = e^{-x}$$

$$\text{Exp}(2) \text{ PDF: } q(x) = 2e^{-2x}$$

This shows $\text{Exp}(2)$ is closer in shape to the target function

(iii) Explaining the Link between (ii) and (i)

from (ii), $q(x) = \text{Exp}(2)$ look similar to $f(x)p(x) = e^{\cos(x) - 2x}$

Monte Carlo:

$$MC = \frac{1}{n} \sum_{i=1}^n e^{\cos(x_i) - 2x_i}, \quad x_i \sim \text{Exp}(1)$$

this calculation makes the estimation with high variance

Importance Sampling:

$$IS = \frac{1}{n} \sum_{i=1}^n \frac{e^{-x_i}}{2e^{-2x_i}} \cdot e^{\cos(x_i) - 2x_i} = \frac{1}{2n} \sum_{i=1}^n e^{\cos(x_i)}, \quad x_i \sim \text{Exp}(2)$$

this calculation has none exponential decay (e^{-x} or e^{-2x})
only dealing with $e^{\cos(x)}$, which takes the
calculation stable.

In conclusion $\text{Exp}(2)$ puts more sample points near small values of x ,
where $e^{\cos(x) - 2x}$ is largest, which makes the importance
sampling method with stable values and smaller variance.

Question 3

(i) Does the time series show clear seasonal behaviour?

"decompose (temperature)"

→ Seasonal component shows repeating cycles of 12 months

Yes, the time series clearly exhibits seasonal behaviour.

This can be found through the consistent seasonal pattern in the decomposed plot.

(ii) Which model is more accurate: Holt-Winters or ARIMA?

$$MSE = \frac{1}{n} \sum_{i=1}^{48} (\text{Actual} - \text{Fitted})^2$$

"fitted (hw-model)" for Holt-Winters

"fitted (arima-model)" for ARIMA

Model	MSE	Accuracy
Holt-Winters	1.84	Moderate
ARIMA	0.97	Higher

In conclusion, ARIMA demonstrates higher prediction accuracy due to a lower MSE. This shows ARIMA has a better ability to capture the underlying pattern in historical data.